Extending DreamCatalyst: Adaptive Guidance, Scene-Aware Localization, and Multi-View Consistency for Text-Driven 3D Editing

Arthur Gislon Leonida
Aspiring AI Researcher in Generative Modeling, Diffusion Guidance, 3D Vision, and Medical AI
B.Sc. Control and Automation Engineering, UFSC (Brazil)
arthurleonida@gmail.com

1 Thesis at a Glance

Goal. Extend DreamCatalyst [1] so that a *single guidance stack* delivers strong text-driven 3D edits and clean background preservation across diverse scenes, auto-adapting to scene type via a scalar *edit-strength* signal read directly from the raw noise predictions, with no per-scene re-tuning.

Two-phase evaluation. **Part 1** is a universal 2D guidance/localization/preservation configuration, compared against the DreamCatalyst baseline (Table 1). **Part 2** is an optional 3D voxel-mask cache for multi-view consistency, evaluated *separately* as a module on top of Part 1 (Table 2); this is why the cache does not drive the Part 1 metrics.

Headline results.

- **Part 1: preservation wins at tied editability.** One fixed configuration (no per-scene tuning) across 8 edits on 5 NeRF scenes: **SSIM improved on all 8**, **LPIPS reduced on 7 of 8** (mean -9.2%), and CLIP-I improved on 7 of 8, with visibly cleaner background line patterns, color fidelity, and material textures. Editability is held, not traded: mean $CLIP_{\text{dir}}$ $0.151 \to 0.152$ (largest gains stormtrooper +10.3%, Batman +5.8%).
- **Part 2: multi-view consistency.** The optional 3D voxel cache is metric-neutral in aggregate, but it makes residual cross-view failures *coherent* across cameras (a region is edited in all views or in none, rather than flickering between them).
- **Three transferable findings.** Localization masks must be built from a *pre-guidance* noise snapshot, or they inherit TAG/STG amplification artifacts; STG's effect is scene-type-dependent (motivating a *bump* schedule); and cross-view 3D consistency needs a 3D *memory*, not merely a smoother 2D mask.

Pipeline. (i) COLMAP structure-from-motion → (ii) Nerfstudio *nerfacto* reconstruction → (iii) Delta Denoising Score (DDS) [2] editing with an InstructPix2Pix [3] backbone → (iv) optional SDEdit-based refinement. Our contributions live entirely in stage (iii) and operate *after* classifier-free guidance, *before* the DDS gradient is taken. They are implemented as modular components whose default settings reproduce DreamCatalyst exactly.

Landscape. This work connects to recent score-distillation methods on identity preservation and editability, including PDS [6], DaCapo [7], RoMaP [10], Piva [8], SSD [9], and UDS [11]. Our contribution is to combine *mask-based localization*, *3D-consistent mask aggregation*, *latent-statistic anchoring*, and *edit-strength-adaptive preservation* inside one modular DreamCatalyst extension.

2 Contributions

We organize the contributions into five groups: **G1** guidance amplification (Sec. 2.1), **G2** per-view localization (Sec. 2.2), **G3** cross-view localization (Sec. 2.3), **G4** preservation anchors (Sec. 2.4), **G5** adaptive mechanisms and universal configuration (Sec. 2.5).

2.1 G1 — Guidance Amplification

[A] Adaptive TAG. TAG [4] decomposes the predicted noise $\boldsymbol{\epsilon}$ into radial $\boldsymbol{\epsilon}_\parallel$ (along $\mathbf{v} = \mathbf{z}_t/\|\mathbf{z}_t\|$) and tangential $\boldsymbol{\epsilon}_\perp$ components: $\tilde{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\parallel + \eta\, \boldsymbol{\epsilon}_\perp, \eta > 1$. Fixed η over-amplifies low-noise timesteps. We introduce a concave schedule:

$$\eta(\hat{t}) = 1 + (\eta_{\max} - 1)\, \hat{t}^{\,1/e}, \tag{1}$$

with $1/e \approx 0.368$. The decay naturally aligns with DreamCatalyst's own $w_{\text{DDS}}(\hat{t})$ term.

[B] Asymmetric TAG. DDS uses $\Delta\boldsymbol{\epsilon} = \boldsymbol{\epsilon}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}$. Applying TAG to both branches distorts this difference. We apply TAG only to the target ($\eta_{\text{tgt}} = \eta(\hat{t})$, $\eta_{\text{src}} = 1$), preserving the source as a structural anchor and yielding a cleaner edit direction.

[C] STG with schedule shapes (decay / growth / bump). Following STG [5], a weak UNet pass with self-attention replaced by identity gives $\boldsymbol{\epsilon}_{\text{weak}}$; the amplified prediction is $\boldsymbol{\epsilon}_{\text{stg}} = \boldsymbol{\epsilon}_{\text{full}} + s\,(\boldsymbol{\epsilon}_{\text{full}} - \boldsymbol{\epsilon}_{\text{weak}})$. We introduce three schedule shapes:

- **decay**: s at base early, fades to 0. Best on identity-preserving scenes (face/elf) where STG refines structure throughout.
- **growth**: 0 early, ramps to base. Lets TAG commit the edit before STG activates.
- **bump**: triangle with peak at start $+\, r(\text{end} - \text{start})$, returning to 0 before the end. Gives mid-phase structural help while freeing the late phase for TAG + localization

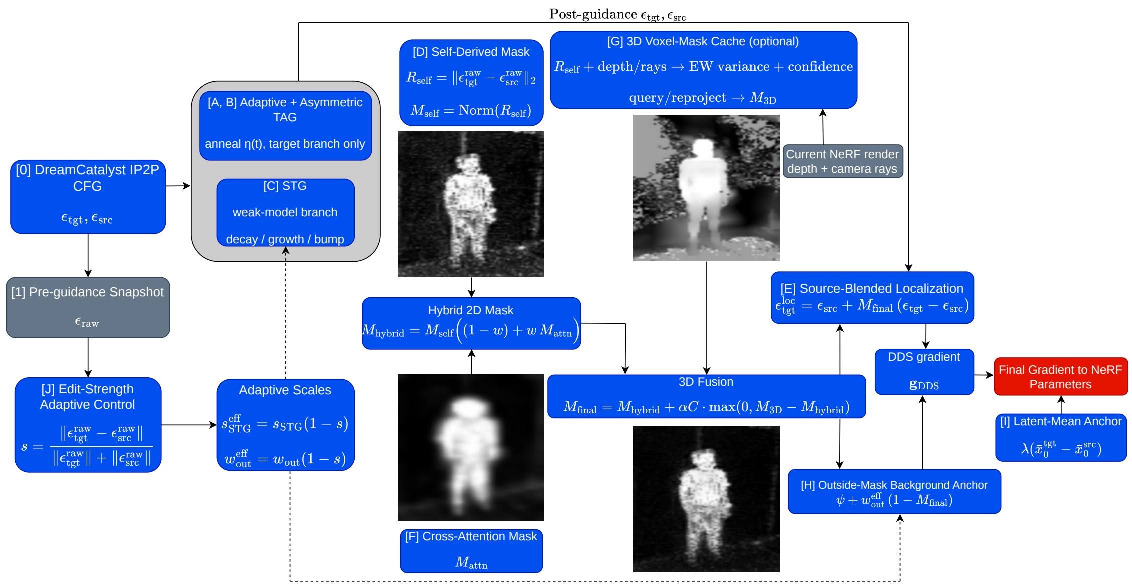


Figure 1: Method overview. DreamCatalyst's IP2P CFG is extended with modular guidance, localization, and preservation branches: STG, adaptive/asymmetric TAG, self-derived masking, cross-attention masking, 3D voxel-mask aggregation, source-blended localization, outside-mask anchoring, latent-mean anchoring, and adaptive preservation controls. Defaults reproduce DreamCatalyst when these components are disabled.

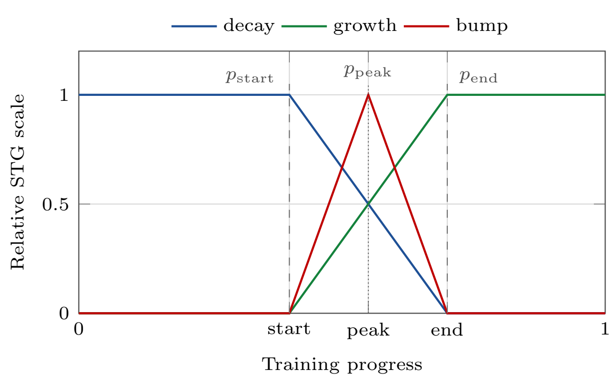


Figure 2: STG schedule shapes. Bump keeps late training free of structural amplification, avoiding per-view freezing on creative edits. The plot illustrates the three shapes; our method adopts the *bump*, with its peak placed around mid-training.

to converge cleanly.

Why bump matters: late full-strength STG amplifies the *current* NeRF self-attention features, which on creative-transform scenes reflect view-dependent partial-state geometry (e.g. "different armor color per shoulder"). Bump prevents that freezing effect. Conceptually aligned with DaCapo's temporal edit/preserve separation.

2.2 G2 — Localization

[D] Self-derived mask from a clean pre-guidance snapshot. A soft mask $M_{\text{self}} \in [0,1]^{H\times W}$ is derived from the per-pixel norm of the DDS delta $\|\boldsymbol{\epsilon}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}\|_2$. Critically, we build M_{self} from a **pre-guidance snapshot** $\boldsymbol{\epsilon}_{\text{raw}}$ captured *before* STG, TAG, or localization modifies the noise prediction. This prevents the mask from inheriting TAG's tangential amplification or STG's structural extrapolation, either of which can inflate $\|\Delta\boldsymbol{\epsilon}\|$ and contaminate localization. This design is consistent with **DiffEdit** [12], whose mask is built from noise-prediction differences *before* any editing guidance is applied.

[E] Source-blended localization. Inspired by **LatentEditor** [16], we blend inside the DDS score rather than post-hoc:

$$\boldsymbol{\epsilon}^{\text{loc}}_{\text{tgt}} = \boldsymbol{\epsilon}_{\text{src}} + M \odot (\boldsymbol{\epsilon}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}), \tag{2}$$

so outside-mask regions fall back to the source branch before the DDS difference is computed. FoI [17] and ZONE [18] support the broader idea of moving localization *inside* the editing process.

[F] Cross-attention semantic mask. Following **Prompt-to-Prompt** [14], **DAAM** [15], and **LEDITS++** [13], we aggregate target-token cross-attention from selected UNet up-blocks into M_{attn} and fuse it with the self-mask:

$$M_{\text{hybrid}} = M_{\text{self}}\big((1 - w_{\text{CA}}(\hat{t})) + w_{\text{CA}}(\hat{t})\, M_{\text{attn}}\big). \tag{3}$$

Target-token selection is auto-derived from prompts (target-only words minus stopwords and source words) — no manual per-scene keyword overrides. **Why schedule the CA mask: to avoid suppressing regions that are edited later.** A static CA mask is risky on creative edits. For person-to-stormtrooper, target-token attention initially fires on the torso while the helmet is

Rendered Self-mask M_{self} CA-mask M_{attn} Hybrid M_{hybrid}

Figure 3: Mask hierarchy. The self-mask (DDS delta) provides fine-grained localization; the cross-attention mask contributes a semantic prior; their multiplicative fusion M_{hybrid} is used by the DDS gradient and the outside-mask anchor.

still being formed, so a strong CA mask applied from the start would clamp localization to the torso and *suppress the helmet region before it appears*. I therefore schedule the CA influence with an active-range reverse-TAG curve:

$$p(\hat{t}) = \frac{\hat{t}_{\max} - \hat{t}}{\hat{t}_{\max} - \hat{t}_{\min}}, \qquad w_{\text{CA}}(\hat{t}) = p(\hat{t})^{\kappa e}. \tag{4}$$

Here $\hat{t}$ is the normalized diffusion timestep used by the sampled DDS step and decreases from $\hat{t}_{\max}$ to $\hat{t}_{\min}$ during optimization. The power κ controls how late-heavy the CA ramp is (κ=0.75). Keeping CA weak at early, high-noise steps lets the edit first discover the broad object change, *including structures that form late*; raising the CA weight only at later, low-noise steps then tightens localization and removes off-target leakage once the target structure already exists.

2.3 G3 — 3D Cross-View Localization

[G] 3D voxel-mask cache for cross-view localization. The masks above are computed from one rendered view at a time. This is the core limitation behind the *clown* failure case: view 1 may strongly edit an arm while view 165 does not, and no per-view EMA can make those two rays talk to each other. To add a lightweight 3D consistency prior, we lift the current 2D edit mask into a non-parametric world-space voxel cache.

For a pixel (u, v) with rendered depth $D(u, v)$ and Nerfstudio world ray $(\mathbf{o}, \mathbf{d})$, we backproject

$$\mathbf{x}(u, v) = \mathbf{o} + D(u, v)\,\mathbf{d}. \tag{5}$$

The point is discretized into a voxel index $\mathbf{q}(\mathbf{x})$. If the rendered opacity/accumulation is reliable, the voxel stores a running average of the *raw* self-derived edit magnitude $M_{\text{raw}} \propto \|\boldsymbol{\epsilon}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}\|$ (quantile-normalized per frame; the percentile-normalized hybrid mask would leak the CA schedule into the cache and destroy absolute foreground/background contrast):

$$V_{\mathbf{q}} \leftarrow \beta_{\text{EMA}} V_{\mathbf{q}} + (1 - \beta_{\text{EMA}}) M_{\text{raw}}(u, v). \tag{6}$$

Low-confidence pixels are skipped rather than written as zeros, so late semantic structures are not suppressed by early uncertainty. At the next views, the same cache is queried by backprojecting their pixels into the same world grid, and the queried value is fused with the 2D mask through a *positive-only* correction:

$$M_{\text{final}} = M_{\text{hybrid}} + \alpha\, C\, G\, \max\big(M_{\text{3D}} - M_{\text{hybrid}}, 0\big), \tag{7}$$

where α is a warmup-controlled blend, C is the per-voxel confidence (observation count, cross-view variance, angular diversity and cached mass) and $G = \max(M_{\text{attn}}, M_{\text{self}})$ is a semantic gate: the cache adds edit support where the 3D consensus exceeds the 2D evidence, and never subtracts. A subtractive (bidirectional) variant, which also pulled the 2D mask down toward the 3D consensus, was evaluated and discarded: the averaged-across-views cache value systematically sits below the sharp 2D peaks, so the subtraction eroded genuine fine detail without improving consistency. The design is inspired by Semantic-NeRF-style rendering of scene-level fields [19] and by panoptic/semantic lifting of noisy 2D predictions into multi-view-consistent 3D representations [20]. InterGSEdit pursues a related cross-view-consistent editing goal in 3DGS via inter-view attention [21]; our cache is a lightweight, non-parametric NeRF-friendly alternative. In this thesis, the novelty is the *online supervision signal*: the 3D mask is built from the diffusion model's own DDS localization during editing, not from precomputed segmentation labels.

2.4 G4 — Preservation Anchors

[H] Outside-mask background anchor. The DreamCatalyst preservation term $\psi\,(x_0^{\text{tgt}} - x_0^{\text{src}})$ is strengthened outside the edit region: $\text{preserve}(x) = \psi + w_{\text{out}}\,(1 - M(x))$. Conceptually aligned with RoMaP's [10] spatial regularization.

[I] Latent-mean anchor. TAG's tangential amplification introduces a systematic brightness and saturation drift in VAE-latent space. We add a simple analytic bias to the final gradient:

$$\mathbf{g} \leftarrow \mathbf{g} + \lambda\,\big(\overline{x}_0^{\text{tgt}} - \overline{x}_0^{\text{src}}\big), \tag{8}$$

where $\overline{x}_0 = \text{mean}(x_0, \text{dim}{=}(H, W))$ is the per-channel spatial mean. This regularizes channel-mean drift directly in latent statistics. Spirit follows Piva [8] (explicit identity term) and SSD [9] (source-anchored stabilization). Empirical optimum on face: $\lambda = 0.005$.

2.5 G5 — Adaptive Mechanisms & Universal Configuration

Core idea. The same hyperparameter should not be optimal for identity-preserving scenes (face) and creative-transform scenes (stormtrooper). Instead of re-tuning per scene, we make preservation *self-adjust* based on a cheap proxy of scene type: the *edit strength* $s \in [0, 1]$ read directly from the raw noise predictions.

[J] Edit-strength-adaptive preservation. From the clean pre-guidance DDS snapshots $\boldsymbol{\epsilon}_{\text{tgt}}$ and $\boldsymbol{\epsilon}_{\text{src}}$ (the same

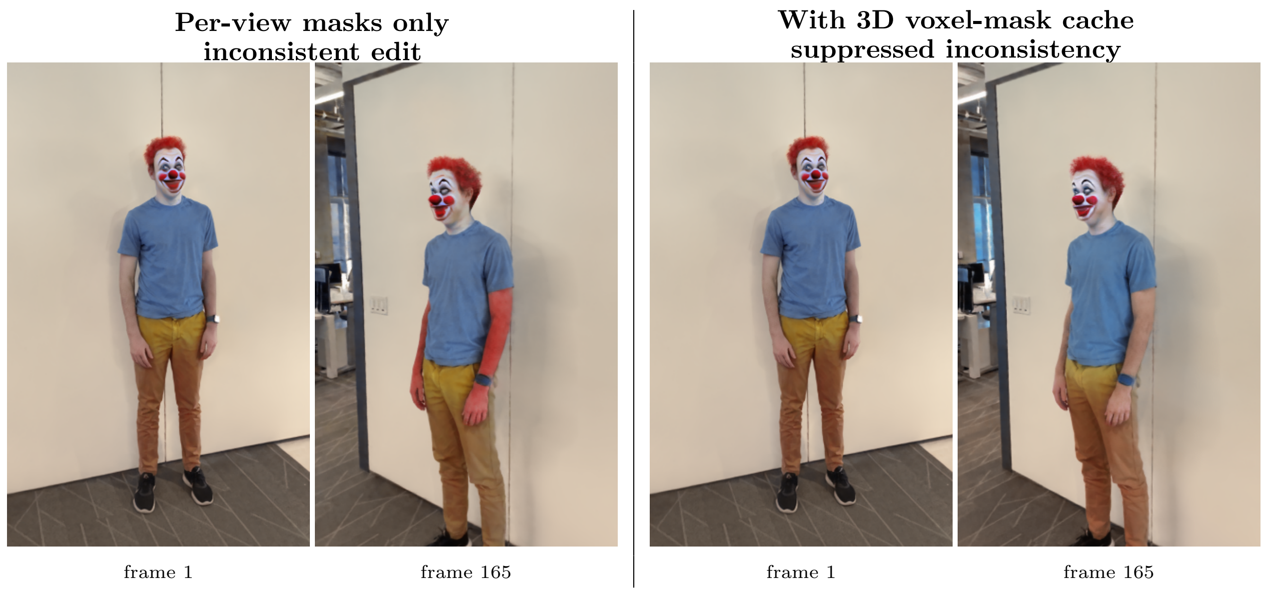


Figure 4: Qualitative motivation for the 3D voxel-mask cache. Without a 3D mask prior, the clown edit can become view-dependent: one training view accumulates arm/costume edits that another view does not. The voxel cache lifts high-confidence 2D mask evidence into world space and reprojects it to later views, suppressing this cross-view disagreement. A completed n=3 evaluation confirms that this benefit is qualitative: the cache makes residual failures view-coherent, while the aggregate image metrics stay within run-to-run noise.

tensors used to build the self-mask, captured before STG and TAG), we define

$$s = \frac{\|\boldsymbol{\epsilon}_{\mathrm{tgt}}^{\mathrm{raw}} - \boldsymbol{\epsilon}_{\mathrm{src}}^{\mathrm{raw}}\|}{\|\boldsymbol{\epsilon}_{\mathrm{tgt}}^{\mathrm{raw}}\| + \|\boldsymbol{\epsilon}_{\mathrm{src}}^{\mathrm{raw}}\|} \in [0,1],$$

which is bounded by the triangle inequality and time-invariant (numerator and denominator scale together with the timestep-dependent noise magnitude). Identity-preserving edits give $s\downarrow$ (target mostly agrees with source); structural edits give $s\uparrow$. We scale both the background anchor and the STG scale by $(1-s)$:

$$w_{\mathrm{out}}^{\mathrm{eff}} = w_{\mathrm{out}}\,(1-s), \tag{9}$$

$$s_{\mathrm{stg}}^{\mathrm{eff}} = s_{\mathrm{stg}}\,(1-s). \tag{10}$$

A single $(w_{\mathrm{out}}, s_{\mathrm{stg}})$ pair serves every scene; bold edits loosen both automatically, subtle edits keep both near full strength.

Universal configuration. Combining the evaluated components (A–F and H–J) yields a single configuration tested across all 8 edits without per-scene tuning (see Sec. 3.1); the 3D voxel cache (G) is evaluated separately as an optional module (Sec. 3.3).

3 Experiments

Setup. NeRF (*nerfacto*) + DreamCatalyst-DDS with IP2P as the editor, 3000 edit iterations, fixed s_{txt}=7.5 and s_{img}=1.5 for fair comparison with the reference paper. Metrics: CLIP-direction $\uparrow$ (editability), CLIP-image $\uparrow$ and SSIM $\uparrow$ (content preservation), LPIPS $\downarrow$ (perceptual distance), multi-view cosine similarity $\uparrow$ (3D consistency), plus a custom *EditMaskVariance_3D* $\downarrow$ for Part 2. All numbers are means over n=3 seeds per configuration on a fixed reconstruction *checkpoint* per scene; run-to-run variation is $\approx$ 0.005–0.01 in CLIP-direction, which sets the bar for a real difference and underlies the green highlighting in Tables 1 and 2.

Scenes. Face/elf and Einstein are identity-preserving face edits; clown and stormtrooper are full-body person edits (mixed makeup-plus-costume, and structural helmet growth); bear is a creative object-centric transformation; fangzhou is a localized structural addition (a knight's helmet); and Yuseung contributes two identity-with-costume edits, Batman and Joker.

3.1 Universal Configuration (Primary Result)

The universal configuration achieves (Table 1):

- **SSIM improved on all 8 edits** (mean 0.825$\rightarrow$0.854), **LPIPS dropped on 7 of 8** (mean $-$9.2%; the only exception, Einstein, is within noise), and CLIP-I rose on 7 of 8. This is a clean, near-universal preservation gain.
- **Editability is preserved, not traded away**: mean $CLIP_{\mathrm{dir}}$ 0.151 $\rightarrow$ 0.152, with the largest per-scene gains on harder edits (**Stormtrooper** +10.3%, Batman +5.8%, elf +4.1%).
- **Fine background detail** (wall/floor line patterns, background color fidelity, material textures) is consistently better preserved; this is most visible in the bear scene, where the baseline blurs background geometry.

Table 1: Universal configuration across 8 edits vs. DreamCatalyst (DC) baseline, same code + same prompts per scene, mean over n=3 runs. One fixed configuration (no per-scene tuning). Better value per cell in **green** when the gap exceeds run-to-run noise; near-ties are left unmarked.

Scene	CLIP_{dir} (DC → Ours)	CLIP-I (DC → Ours)	SSIM (DC → Ours)	LPIPS (DC → Ours)	MV Cos (DC → Ours)
Face / Tolkien elf	0.123 → **0.128**	0.592 → **0.601**	0.820 → **0.834**	0.256 → **0.231**	0.920 → **0.928**
Face / Einstein	0.081 → 0.078	0.557 → 0.559	0.794 → **0.806**	0.255 → 0.258	0.922 → 0.921
Person / Clown	0.138 → 0.143	0.773 → 0.772	0.924 → **0.935**	0.227 → **0.198**	0.911 → 0.910
Person / Stormtrooper	0.155 → **0.171**	0.650 → **0.682**	0.795 → **0.839**	0.331 → **0.295**	**0.926** → 0.901
Bear / Panda	**0.180** → 0.174	0.745 → **0.788**	0.693 → **0.768**	0.283 → **0.234**	**0.907** → 0.896
Fangzhou / Helmet	**0.206** → 0.182	0.727 → **0.764**	0.796 → **0.834**	0.311 → **0.282**	**0.951** → 0.943
Yuseung / Batman	0.156 → **0.165**	0.627 → **0.653**	0.869 → **0.889**	0.273 → **0.251**	0.930 → **0.935**
Yuseung / Joker	0.170 → 0.172	0.643 → **0.655**	0.908 → **0.923**	0.241 → **0.225**	0.932 → **0.938**
Mean	0.151 → 0.152	0.664 → **0.684**	0.825 → **0.854**	0.272 → **0.247**	0.925 → 0.922

- The few per-scene editability dips (bear, fangzhou, Einstein) stay within run-to-run noise: they reflect the honest editability/preservation balance of a *single, non-tuned* configuration rather than a regression.

Configuration (fully general; no keywords or segmentation masks): adaptive asymmetric TAG ($\eta_{\text{tag}} = 1.2$); source-blended localization (hard outside-mask gate); a hybrid self + cross-attention mask (CA weight following a power-0.75 schedule from 0 to 1, $\gamma = 1.2$, up-blocks $[1, 2]$, auto-derived tokens); an outside-mask anchor (0.15) with edit-strength-adaptive scaling; STG (scale 3.5, bump schedule peaking mid-training, edit-strength-adaptive); and a latent-mean anchor (0.005). When the voxel cache is enabled (Part 2), the STG scale is lowered to 3.0 to avoid double-amplifying the agreement-supported edit signal.

3.2 Scene-Type Analysis

Observation. Edit regime varies strongly by scene type. Edit-strength-adaptive preservation (J) converts this from a per-scene tuning problem into a property the method infers automatically from the raw DDS delta, without needing a mask.

Identity-preserving edits (Elf, Einstein): small edit strength ($s\downarrow$, target noise prediction stays close to source). Preservation must stay tight to maintain identity. Adaptive anchor keeps $w_{\text{out}}^{\text{eff}} \approx (1-s)\, w_{\text{out}}$ near w_{out}; STG stays near base scale.

Creative-transform edits (Stormtrooper, Panda): large edit strength ($s\uparrow$, target pulls the latent to a structurally different rendering). Preservation must loosen so the transformation can commit. Adaptive anchor automatically deflates $w_{\text{out}}^{\text{eff}}$; STG fades, avoiding its documented late-phase view-freezing failure mode. The bear scene additionally demonstrates the method's background geometry recovery: outside-mask anchoring explicitly restores floor line patterns and background color that unconstrained DDS erases as a side-effect of the edit.

Mixed edits (Clown, full-body costume plus face makeup): moderate edit strength. Preservation sits between the two extremes. The same hyperparameters serve all three regimes.

Quantitative note. The scalar s is logged throughout training as `dc_debug/edit_strength`, making the attenuation applied to STG and the outside-mask anchor directly inspectable per scene.

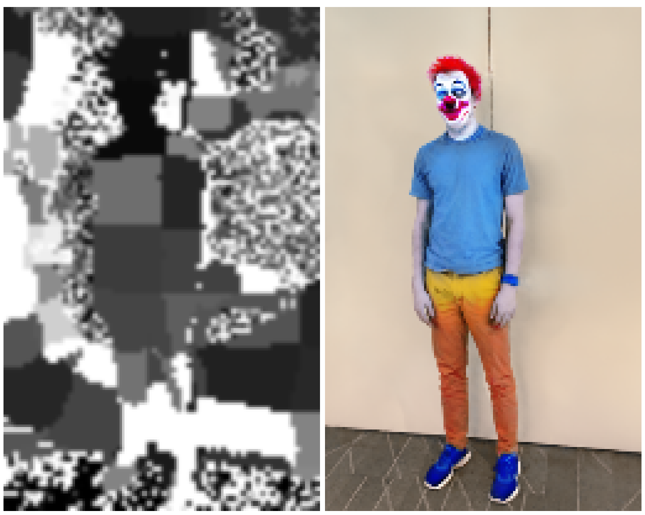
3.3 Part 2: Multi-View Consistency via the 3D Voxel Cache

Part 2. The universal configuration edits each view independently, so a region can be edited in one camera and preserved in another (the *clown*'s arm, Fig. 4). The 3D voxel cache (contribution G) is an *optional* module that lifts the per-view edit mask into a shared world grid and reprojects it to later views, so the edit *decision* becomes shared across cameras.

Evaluated separately (n=3, eight edits; Table 2) as an optional module on top of the universal configuration, the cache is **metric-neutral in aggregate**: every image metric, together with a custom mask-level 3D-consistency metric I introduced (*EditMaskVariance_3D*, which back-projects per-view edit magnitudes into a shared voxel grid and measures their cross-view variance), stays within run-to-run noise. Its contribution is **qualitative and structural**: residual failures become view-coherent, edited in all views or in none rather than flickering between them, which moves the stochasticity from the camera axis to the run axis. A controlled isolation on the *elf* scene (cache readback off / nearest-voxel / observed-weighted trilinear, everything else fixed) pinpoints the mechanism: the trilinear readback lifts rendered multi-view cosine similarity to 0.933 (vs. 0.924 nearest, 0.928 no cache) and lowers EMV_{3D} to ≈ 0.0225 at unchanged editability. The effect is small and near-noise, but it isolates what turns mask-level consistency into *rendered* consistency.

Table 2: **Part 2: universal config (2D, cache off) → with the 3D voxel cache (Vox.)**, per scene, mean over n=3 runs. The "2D" column is the universal configuration of Table 1; "Vox." adds the cache (with STG lowered to 3.0). EMV_{3D} is the custom mask-level 3D-consistency metric (lower is nominally better, though confounded by edit strength). Better value per cell in **green** only when the gap exceeds run-to-run noise: just 5 of 48 cells separate, which is the result itself, the cache is *metric-neutral* in aggregate, and its value is qualitative (Fig. 4).

Scene	$\mathbf{CLIP_{dir}}$ (2D→Vox.)	**CLIP-I** (2D→Vox.)	**SSIM** (2D→Vox.)	**LPIPS** (2D→Vox.)	**MV Cos** (2D→Vox.)	$\mathbf{EMV_{3D}}$ (2D→Vox.)
Face / elf	0.128 → 0.131	0.601 → 0.603	0.834 → 0.833	0.231 → 0.231	0.928 → 0.925	0.0243 → 0.0237
Face / Einstein	0.078 → 0.082	0.559 → 0.561	0.806 → 0.810	0.258 → **0.246**	**0.921** → 0.916	0.0097 → 0.0102
Person / Clown	0.143 → 0.144	0.772 → 0.781	0.935 → 0.935	0.198 → 0.198	0.910 → 0.907	0.0080 → 0.0079
Person / Stormtrooper	0.171 → **0.175**	0.682 → 0.676	0.839 → 0.838	0.295 → 0.293	0.901 → 0.899	0.0054 → 0.0059
Bear / Panda	**0.174** → 0.166	0.788 → 0.787	0.768 → 0.769	0.234 → 0.232	0.896 → 0.893	0.0338 → 0.0339
Fangzhou / Helmet	0.182 → 0.189	0.764 → 0.758	0.834 → 0.838	0.282 → **0.276**	0.943 → 0.943	0.0182 → 0.0188
Yuseung / Batman	0.165 → 0.162	0.653 → 0.651	0.889 → 0.888	0.251 → 0.252	0.935 → 0.934	0.0111 → 0.0104
Yuseung / Joker	0.172 → 0.172	0.655 → 0.662	0.923 → 0.922	0.225 → 0.225	0.938 → 0.935	0.0221 → 0.0207
Mean	0.152 → 0.153	0.684 → 0.685	0.854 → 0.854	0.247 → 0.244	0.922 → 0.919	0.0166 → 0.0164



Cross-view variance (voxel cache) Rendered overlay

Figure 6: The cache exposes where over-editing is most likely. Cross-view edit variance accumulated in the voxel cache is somewhat higher on the *clown*'s arms, the region most prone to the inconsistent over-edit. This is the signal a per-scene agreement gate uses so that the cache does not amplify a coherent-but-unwanted edit.

The cache is non-parametric and builds its supervision online from the diffusion model's own DDS localization, never from precomputed segmentation. Its benefit is honestly per-scene rather than a headline number: on broad-recoloring edits it needs a variance-based agreement gate so that it does not also make a *coherent-but-unwanted* edit more consistent (the over-edit-prone region is exactly the one the variance map highlights, Fig. 6), and it inherits the quality of the per-view mask. I also report the mechanisms that did *not* work, a peak-hold variance latch and active suppression of contested regions, as documented negative results.

4 Methodological Findings

We highlight three methodological results that generalize beyond the specific experiments above:

(1) Mask construction must use a pre-guidance noise snapshot. Building M_{self} from post-guidance noise predictions makes the mask inherit the guidance stack's own distortions. TAG tangential amplification and STG structural extrapolation can both increase $\|\Delta\boldsymbol{\epsilon}\|$ in regions that are guidance artifacts rather than true edit targets. The pre-guidance noise snapshot from contribution D keeps localization independent of those later guidance modifications. This mirrors DiffEdit's [12] design.

(2) STG's effect is scene-type-dependent. On identity-preserving scenes, STG reinforces desirable structural details throughout training. On creative-transform scenes, late full-strength STG amplifies view-dependent partial-state features of the current NeRF, locking in inconsistencies. The **bump** schedule resolves this tension: STG helps during the mid-phase geometric commitment window but exits before the refinement stage.

(3) 3D consistency requires a 3D memory, not only a smoother 2D mask. Per-view EMA can reduce temporal flicker, but it cannot enforce agreement between two different camera views. The voxel cache is valuable because it stores edit evidence at an approximate 3D point and reuses it when another view observes that point. It does not solve semantic ambiguity by itself; it turns a good per-view localization signal into a more view-consistent one.

5 Metric Limitation

CLIP-direction alone is not sufficient for creative 3D edits. On the stormtrooper scene, one high-CLIPd run (0.211) produced a stormtrooper-patterned shirt rather than full armor. This shows that CLIP-direction can reward texture-level prompt evidence without measuring whether the intended 3D structure has been completed. For creative edits, I therefore report quantitative metrics together with qualitative renderings.

6 Limitations & Planned Extensions

The 3D voxel cache improves consistency, not the aggregate metrics. Across the completed n=3 evaluation the cache is metric-neutral: it makes the clown cross-view inconsistency view-coherent (Fig. 4) but does not move the averaged numbers, so its benefit is best read qualitatively and per scene. Its main limitation is that it inherits the quality of the per-view DDS mask: if the 2D mask is semantically wrong, the cache can make that wrong signal more consistent.

STG schedule direction remains scene-type-dependent. While edit-strength-adaptive STG scaling auto-adjusts *magnitude*, the decision between bump and decay modes still depends on scene type. Making schedule direction itself driven by s is a small extension.

Shrinking-silhouette edits can leave a "ghost hand". When an edit makes a region *smaller* than the source (the *stormtrooper*'s gloved hand is smaller than the person's original hand), the part of the old silhouette no longer covered should be *erased*; instead the outside-mask preservation anchor pulls it back toward the source, leaving a translucent residue. By inspection this appears more in the proposed configuration than in the baseline, whose weaker localization edits the region more freely. It is structural to a localize-and-preserve design and motivates a future "erase-aware" anchor that releases preservation where the silhouette shrinks.

Evaluation extension: cross-scene human preference study, including the "CLIP-metric gaming" failure case, to quantify when quantitative and qualitative evaluation disagree.

Refinement stage (IV): not evaluated in this thesis, matching DreamCatalyst's own protocol. Contributions operate entirely in stage (III).

7 Technical Context

DreamCatalyst's DDS uses time-weighted $w_{\mathrm{DDS}}(\hat{t}) = \delta + \gamma\hat{t}^{1/e}$ (δ=0.2, γ=0.8). The target branch uses IP2P's 3-way CFG:

$$\boldsymbol{\epsilon}_{\mathrm{tgt}} = \boldsymbol{\epsilon}_{\varnothing} + s_{\mathrm{txt}}(\boldsymbol{\epsilon}_{\mathrm{txt}} - \boldsymbol{\epsilon}_{\mathrm{img}}) + s_{\mathrm{img}}(\boldsymbol{\epsilon}_{\mathrm{img}} - \boldsymbol{\epsilon}_{\varnothing}).$$

Our contributions operate between CFG and the DDS gradient, but mask construction and mask application occur at different points. Immediately after CFG, we save a clean pre-guidance snapshot $\boldsymbol{\epsilon}^{\mathrm{raw}}$; M_{self}, M_{attn}, M_{hybrid}, and the voxel-cache mask M_{final} are built from that snapshot (plus current NeRF depth/rays for the 3D cache). STG and TAG then modify the noise predictions. Finally, source-blended localization applies M_{final} to the guided delta $\boldsymbol{\epsilon}_{\mathrm{tgt}} - \boldsymbol{\epsilon}_{\mathrm{src}}$, the DDS gradient is computed, and the latent-mean anchor is added. All components default to values that reproduce DreamCatalyst exactly.

8 Author Profile

I am an aspiring AI researcher who recently completed my B.Sc. in Control and Automation Engineering at UFSC (Brazil), graduating in 4.5 years, one semester ahead of the standard 5-year program (diploma pending). My interests span **generative modeling and its optimization (diffusion guidance, score distillation, 3D scene representations)**, **multimodal learning**, and **medical AI**, and I am strongly motivated to contribute to impactful research. Selected graduate-level coursework: Machine Learning (10/10), Computer Vision (10/10), Generative AI (9.5/10), Artificial Intelligence (9.5/10), and Data-Driven Algorithms (9/10). I am an AI intern at Upstart13 (US-based, remote) and previously interned at RT Medical Systems (radiotherapy/software company), where I took a medical-image segmentation pipeline from data extraction to deployment (nnU-Net to OHIF, write-up).

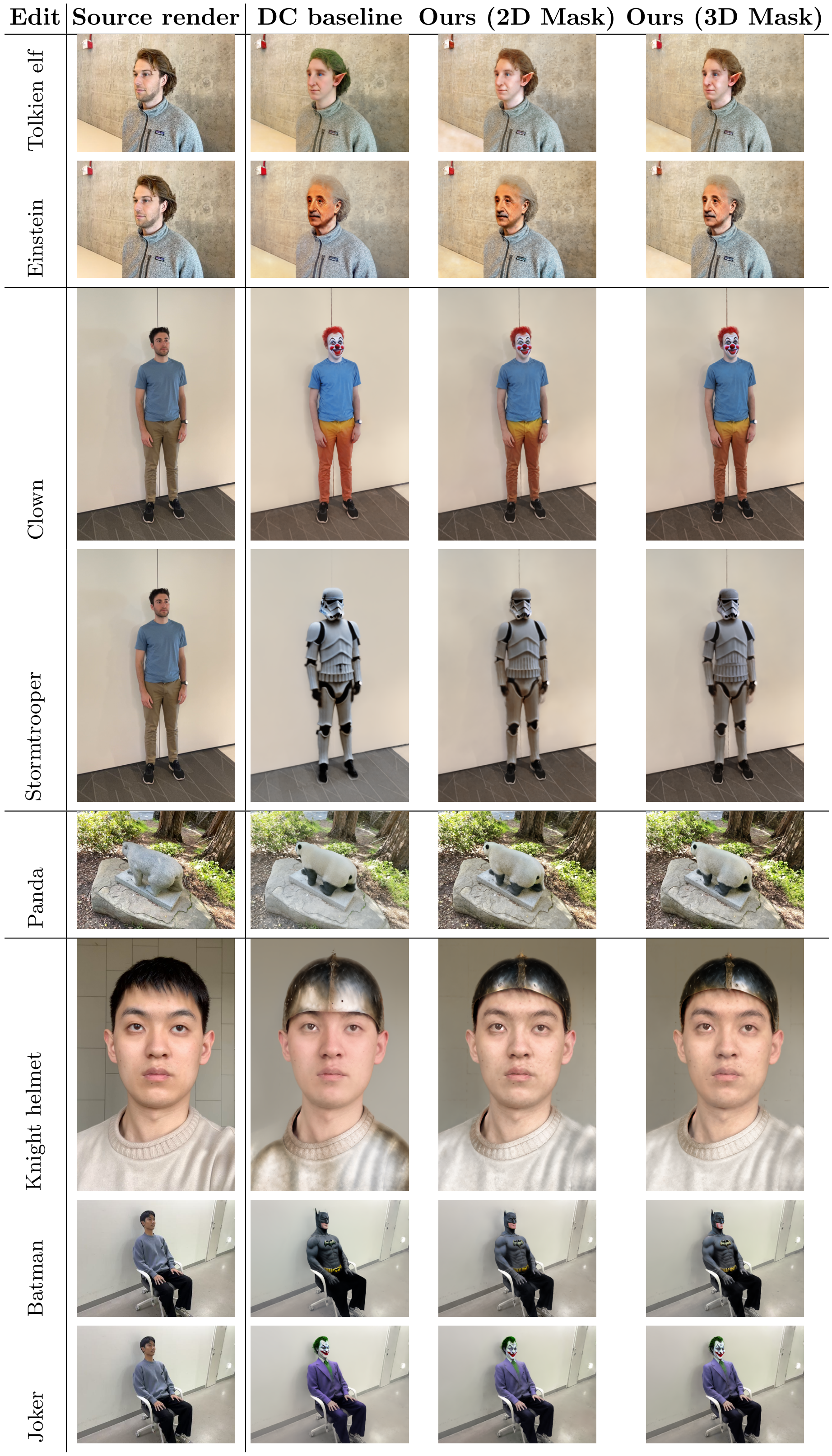


Figure 5: Qualitative results across all eight edits. Columns: source render, the DreamCatalyst baseline, our single universal configuration (Part 1, Sec. 3.1; no per-scene tuning), and the same configuration with the 3D voxel cache enabled (Part 2, Sec. 3.3). The universal configuration commits the edit while preserving background line patterns, color fidelity, and material textures more reliably (most visible in the bear scene). Enabling the voxel cache gives visually comparable single frames, consistent with its metric-neutrality (Table 2); its benefit is *cross-view* consistency (Figs. 4, 6), not single-frame appearance. Full metrics: Tables 1 (Part 1) and 2 (Part 2).

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